

Panelists



Abhijit Chaudhury,
Founder and CEO,
Niral Network



Dr Alok Nath De, CTO,
Samsung India (moderator)



Gnanapriya Chidambaranathan,
AVP and Senior Principal
Architect, Infosys



Dr Inder S. Gopal, CEO,
Indian Urban Data Exchange



Manish Gangey,
SVP and Head, R&D,
Airtel



**Subodh Gajare, Lead
Architect (5G and IoT
Security), Cisco R&D**

can improve upon the operations efficiency, Gnanapriya opines.

The other part is the surveillance, safety, and security solutions for the likes of oil and gas industries. There seems to be lot of opportunities to capture these solutions, but there are challenges in accessing or processing them in real time. “It is important to think on how to bring all the partner ecosystems together in an operations platform that can help in taking solutions forward,” she adds.

Subodh adds that open source has tremendous potential for a testbed. With the open source smart city platforms in Pune, Jaipur, New Bombay, etc we are also trying to do something for the digital freight corridor operation now. We need a registry for digital public goods, a place where all this data can be exchanged at par with the right security levels and context.

“Having a global registry for smart cities, or even just for the 6,000 odd cities in India, would be nice. And that is solely missing as one of the key agenda items, because, if you compare with the world city council

for data, there, they have pretty much squared that circle for a few cities. We are trying to do something similar for India,” Subodh adds.

The 5G is opening the door to multi-fold connectivity, and edge computing may have a significant role in particularly mulching open source. There are two types of edge—user edge and service provider edge. The user edge may be either microcontroller based system or gateway type. There is a focus on creating network infrastructure that may be a private network, especially to enable edge computing. Experts warn that you may be forced to enhance edge computing in future solutions.

Abhijit says, “When we talk about smart factories and rural connectivity, we would need a private network in a lot of cases. We are trying to build an open network operating system for that kind of solution, which needs three components—the radio, the transport, and the core. We can create a concrete edge-upfront core network for the 5G using open source, which is all about collaboration, more com-

petitors coming and collaborating.”

Reiterating the relationship between standards and open source, Inder says, “The way standards get developed is by creating open source reference implementations. There is a kind of cycle between formal standards and implementations. And in many ways, open source drives that standard.”

He goes on to state that a task force in the Indian Standards body looks at the feasibility of 5G, open source for 5G, and in multiple segments like the access and applications space to check on the maturity of open source. “But the objective really is to take it forward and go from a paper study to actually do a set of reference implementations in each of these areas.”

India is known for software programming and many software engineers. But when it comes to open source, we are still at infancy. Subodh feels that time has come for creating something similar for the whole 5G ecosystem, where everyone including a student, academician, or research scholar can contribute. India obviously has the capability, and now it’s just a matter of working with the necessary frameworks for solving something tangible.

“We have huge talent in India. Getting associated with any open source project of interest will help. Students can start with bug fixes and then move and contribute to features. There are a variety of opportunities available at any layer,” suggests Gnanapriya.

Some experts believe that we must develop the culture of contribution rather than looking at reaping immediate benefits. “I think we need to really grow our thought process and nurture this culture, also by busting a lot of myths around it. If you are contributing something to open source, it is not your intellectual property. Let’s build this culture of contributions,” Manish concludes. **EFY**

The article was prepared by Abhinaya Kuzhanthaivel, an assistant editor at EFY, based on the panel discussion held during Open Source Conference 2021.